

Python Assessment

Q.1 Ans:

```
text = "kitten"
```

```
reverse = ""
```

```
for char in text:
```

```
    reverse = char + reverse
```

```
print(reverse)
```

Q.2. Ans:

```
text = "accessories"
```

```
for char in text:
```

```
    if text.count(char) == 1:
```

```
        print(char)
```

```
        Break
```

Q.3. Ans:

```
text = "madam"
```

```
reverse = ""
```

```
for char in text:
```

```
    reverse = char + reverse
```

```
if text == reverse:
```

```
    print(True)
```

```
else:
```

```
    print(False)
```

Q.4 Ans:

```
text = "hello"
```

```
frequency = {}
```

```
for char in text:
```

```
    if char in frequency:
```

```
        frequency[char] += 1
```

```
    else:
```

```
        frequency[char] = 1
```

```
for char, count in frequency.items():  
    print(char + ":" + str(count))
```

Q.5. Ans:

```
text = "programming"  
result = ""
```

```
for char in text:  
    if char not in result:  
        result += char
```

```
print(result)
```

Section B - Lists

Q.6. Ans:

```
numbers = [1, 2, 2, 3, 4, 4]  
result = []
```

```
for num in numbers:  
    if num not in result:  
        result.append(num)
```

```
print(result)
```

Q.7. Ans:

```
numbers = [10, 20, 5, 30, 25]
```

```
largest = numbers[0]  
second_largest = numbers[0]
```

```
for num in numbers:  
    if num > largest:  
        second_largest = largest  
        largest = num  
    elif num > second_largest and num != largest:  
        second_largest = num
```

```
print(second_largest)
```

Q.8. Ans:

```
numbers = [1, 2, 3, 2, 4, 5, 1]
```

```
duplicates = []
```

```
for num in numbers:
```

```
    if numbers.count(num) > 1 and num not in duplicates:
```

```
        duplicates.append(num)
```

```
print(duplicates)
```

Q.9. Ans:

```
numbers = [1, 2, 3, 4, 5]
```

```
k = 2
```

```
k = k % len(numbers)
```

```
result = numbers[-k:] + numbers[:-k]
```

```
print(result)
```

Q.10. Ans:

```
list1 = [1, 2, 3, 4]
```

```
list2 = [3, 4, 5, 6]
```

```
result = []
```

```
for num in list1:
```

```
    if num in list2 and num not in result:
```

```
        result.append(num)
```

```
print(result)
```

Section C - Dictionary

Q.11. Ans:

```
numbers = [1, 2, 2, 3, 3, 3]
```

```
frequency = {}
```

```
for num in numbers:
```

```
    if num in frequency:
```

```
        frequency[num] += 1
    else:
        frequency[num] = 1

print(frequency)
```

Q.12. Ans:

```
data = {"A": 100, "B": 500, "C": 300}
```

```
max_key = ""
max_value = 0
```

```
for key, value in data.items():
    if value > max_value:
        max_value = value
        max_key = key
```

```
print(max_key)
```

Q.13. Ans:

```
data = {"a": 1, "b": 2}
reverse = {}
```

```
for key, value in data.items():
    reverse[value] = key
```

```
print(reverse)
```

Q.14. Ans:

```
d1 = {"a": 1}
d2 = {"b": 2}
```

```
result = {}
```

```
for key, value in d1.items():
    result[key] = value
```

```
for key, value in d2.items():
    result[key] = value
```

```
print(result)
```

Q.15.Ans:

```
sentence = "cooking is good cooking is easy"
```

```
words = sentence.split()
```

```
frequency = {}
```

```
for word in words:
```

```
    if word in frequency:
```

```
        frequency[word] += 1
```

```
    else:
```

```
        frequency[word] = 1
```

```
print(frequency)
```

Section D - Loopa

Q.16.Ans:

```
for i in range(1, 3):
```

```
    for j in range(i):
```

```
        print("*", end="")
```

```
    print()
```

Q.17.Ans:

```
num = 5
```

```
for i in range(1, 11):
```

```
    print(num, "x", i, "=", num * i)
```

Q.18.Ans:

```
num = 5
```

```
factorial = 1
```

```
for i in range(1, num + 1):
```

```
    factorial = factorial * i
```

```
print(factorial)
```

Q.19.Ans:

```
for num in range(2, 101):
    is_prime = True

    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print(num, end=" ")
```

Q.20.Ans:

n = 8

a = 0

b = 1

```
for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

Bonus Questions & Answers

Q.21.Ans:

numbers = [1, 2, 3, 4, 5, 1, 2, 3]

```
for num in numbers:
    if numbers.count(num) == 1:
        print(num)
        break
```

Q.22.Ans:

numbers = [1, 2, 3, 4, 5, 1, 2, 3]

n = 2

count = 0

```
for num in numbers:
    if numbers.count(num) == 1:
        count += 1
```

```
if count == n:  
    print(num)  
    break
```

Q.23.Ans:

```
str1 = "listen"  
str2 = "silent"
```

```
if sorted(str1) == sorted(str2):  
    print(True)  
else:  
    print(False)
```

Q.24.Ans:

```
numbers = [1, 2, 3, 5]
```

```
n = len(numbers) + 1
```

```
total = n * (n + 1) // 2
```

```
missing = total - sum(numbers)
```

```
print(missing)
```

Q.25.Ans:

```
numbers = [1, 2, 2, 3, 3, 3, 4]
```

```
max_count = 0
```

```
top_element = None
```

```
for num in numbers:
```

```
    count = numbers.count(num)
```

```
    if count > max_count:
```

```
        max_count = count
```

```
        top_element = num
```

```
print(top_element)
```

Differences:

Lists	Tuple
Mutable can be changed	Immutable cannot be changed
[]	()
Slower	Faster
Good for data that changes	Good for fixed data

Dictionary	Set
Stores key value pairs	Stores unique values
Uses {key: value}	Uses {value}
Access values using keys	No key-value access
Keys must be unique	All elements must be unique

Is	==
Checks if 2 variables refer to same object	Checks if 2 objects have same value
Identify comparison	Equality comparison

Append VS Extend

Append: adds 1 item

Extend: adds items individually